

FORENSIC INVESTIGATION

Ethan M. Dirks

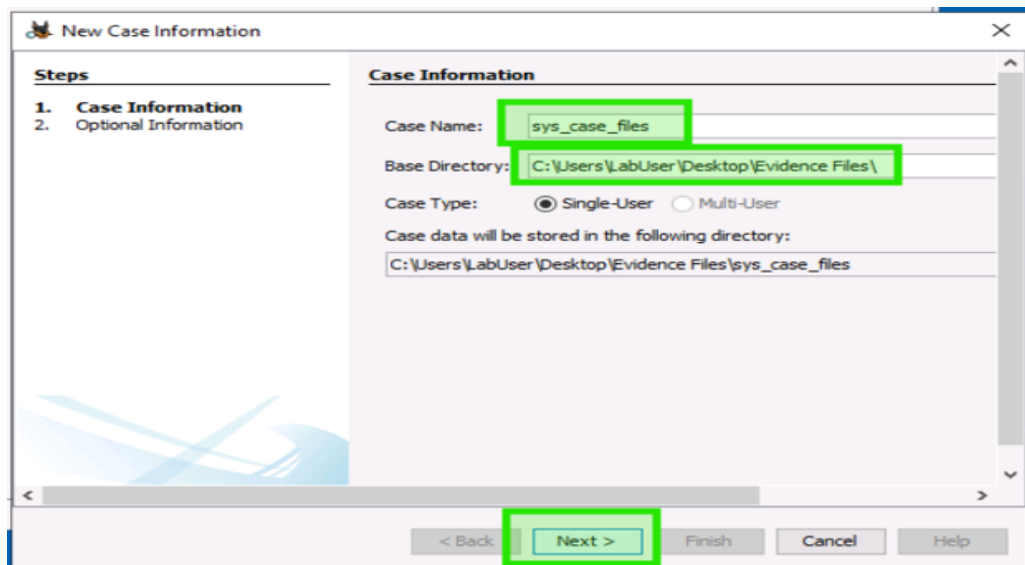
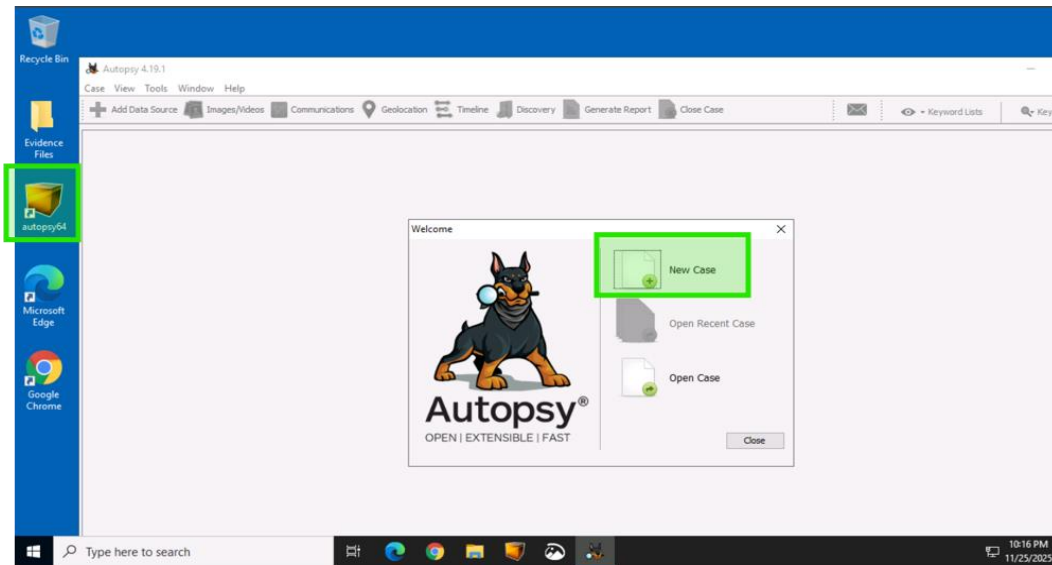
11/23/2025

1. Creating the forensic system case file in Autopsy:

Steps:

1. On Windows menu, double-click “Autopsy64” and wait for the application to run
2. Select “New Case”
3. In “Case Information”, Enter “sys_case_files” for case name and set base drive to “C:\Users\LabUser\Desktop\Evidence Files\” (Option: Use the browse button to find in file directory). Then select “Next”
4. In “Optional Information”, Enter case number as “123456”, and name as “Example User”. Select “Next”
5. Leave “Select Host” as default and select “Next”
6. Select “Disk Image or VM File” as the data source type, then select “Next”
7. Under “path” browse/ enter “C:\Users\LabUser\Desktop\Evidence Files\JSmith_Q1.001” and then select “Next”
8. Leave “Configuration Digest” as default and select "Next"
9. Wait for “Add Data Source” to stop processing. When it’s complete, select finish
10. The case file has now been created! (Consult screenshots for what each step looks like and what the case file menu looks like when all steps are completed)

Screenshots:



New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number: 123456

Examiner

Name: Example User

Phone:

Email:

Notes:

Organization

Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > **Finish** Cancel Help

Add Data Source

Steps

1. **Select Host**
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source

Select Host

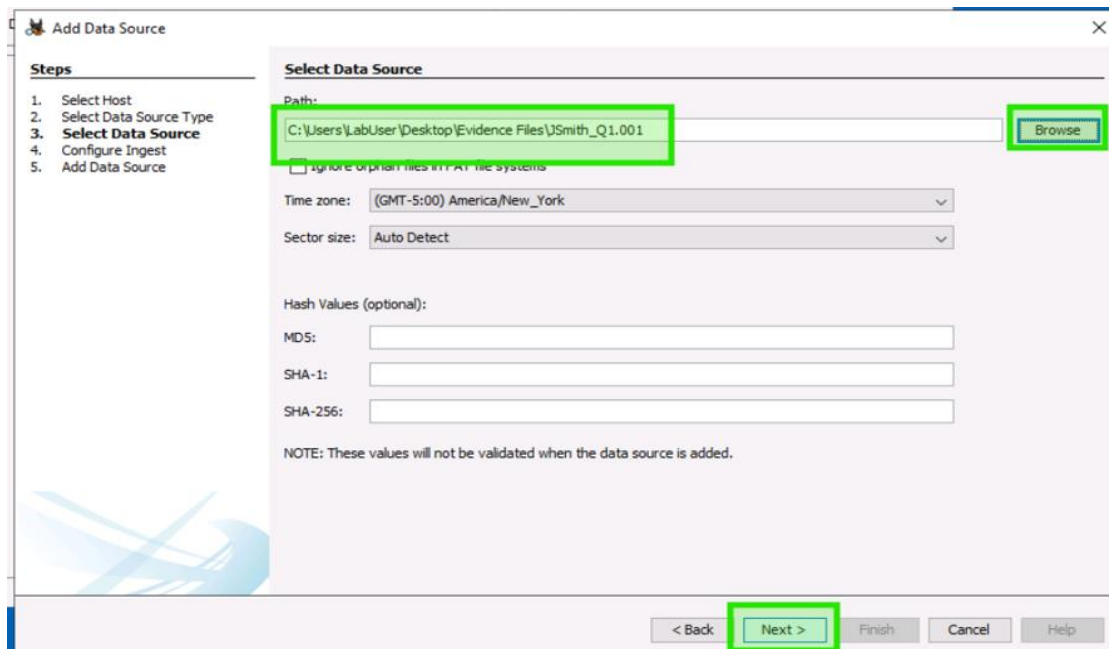
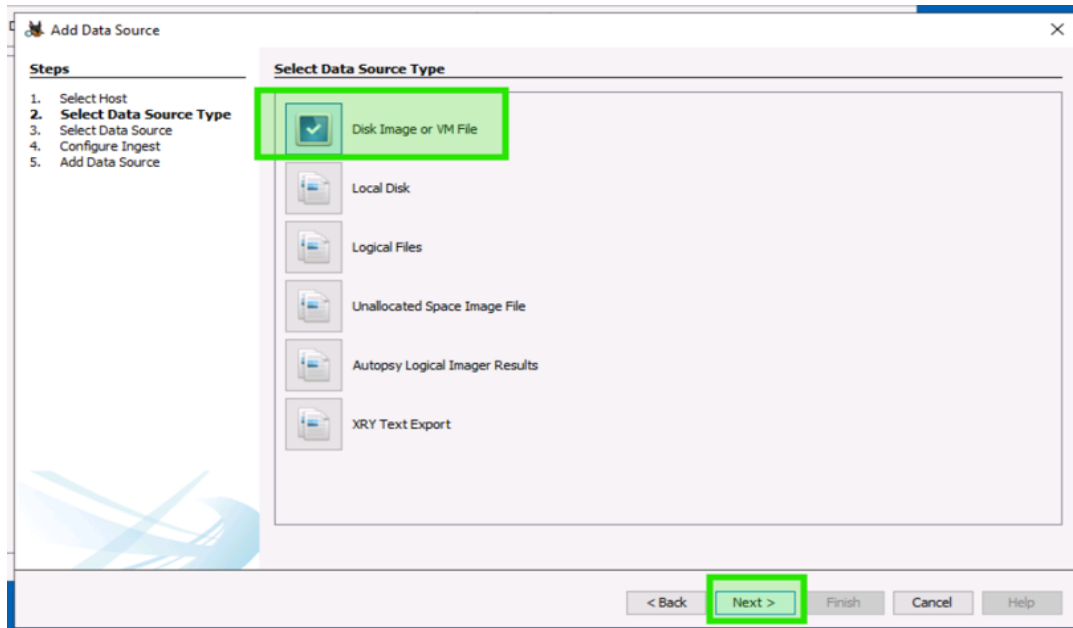
Hosts are used to organize data sources and other data.

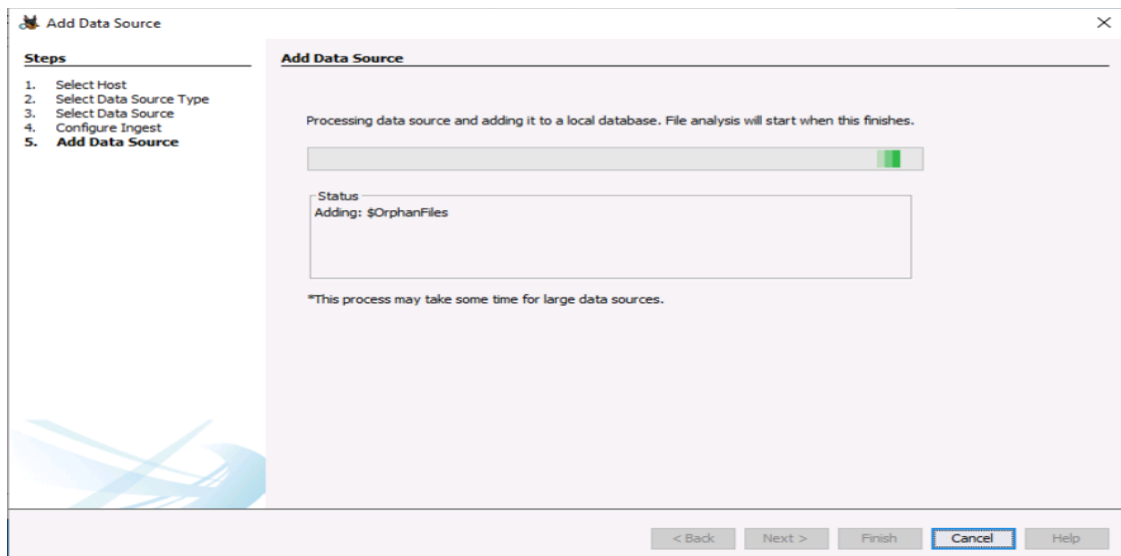
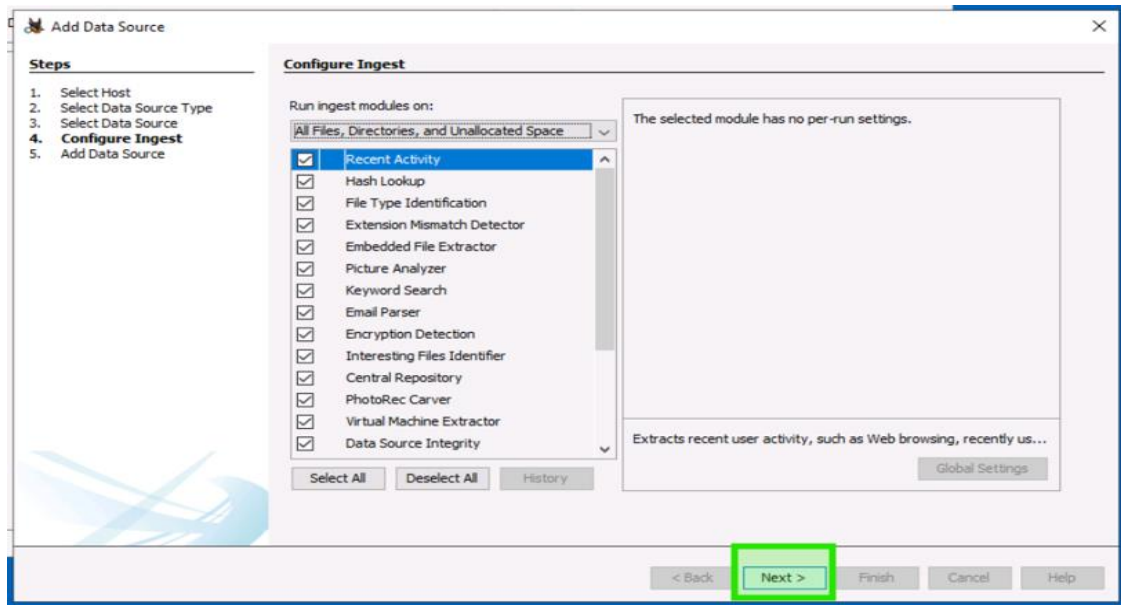
☒ Generate new host name based on data source name

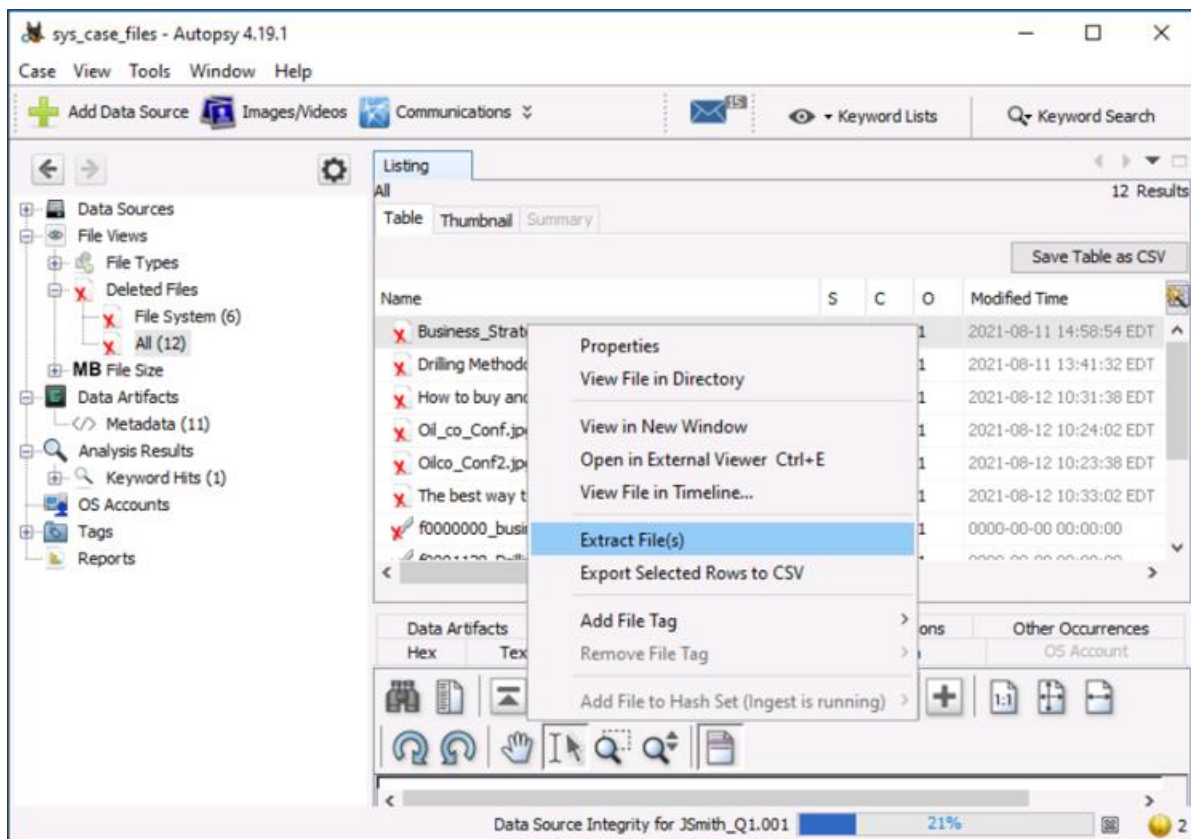
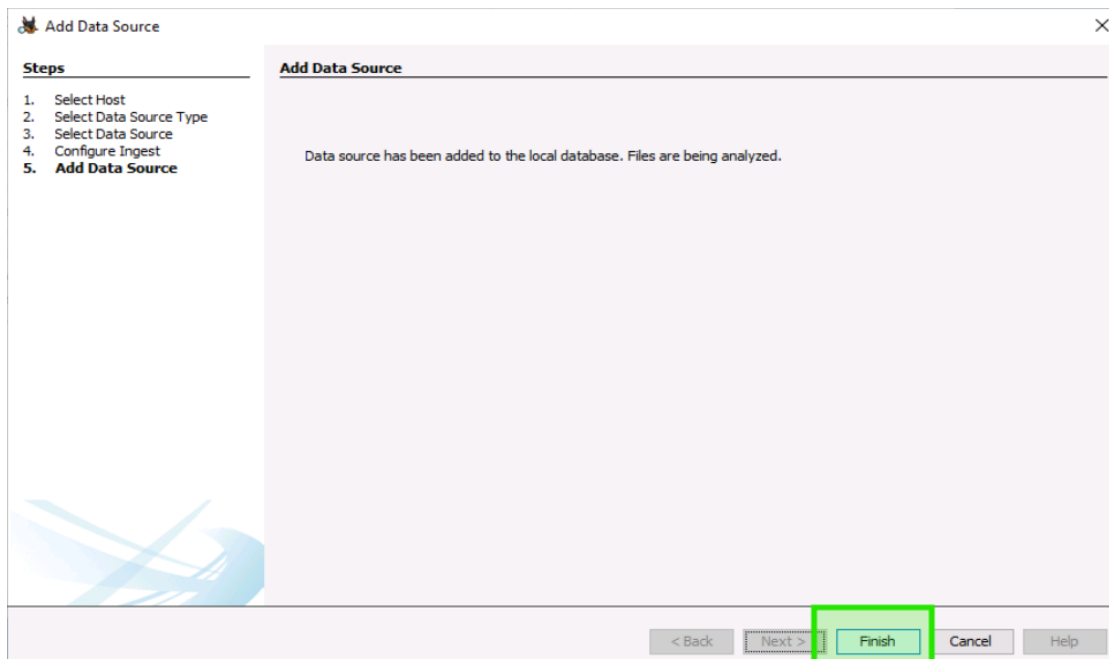
☐ Specify new host name

☐ Use existing host

< Back **Next >** Finish Cancel Help







2. Potential evidence discovery in Autopsy (data files, deleted data files, directories, drive partitions):

Claim made against John Smith: The employee is not authorized to access proprietary information and has violated this restriction by accessing information that was clearly identified as proprietary.

Steps:

1. In the left panel, select “File Views”, followed by “Deleted Files”, then “All”
2. After selecting “All”, for each of the files on the “Listing” tab, review the “Application”, “File Metadata”, and “Data Artifacts” tabs on the lower panel
 - a. The key details to look out for if the files say “Restricted/ Confidential”, John smith is not the original owner of the files, if john smith has made copies of any restricted files, and if information about John Smith’s motives can be found
3. (Optional) For an easier view of metadata select “Metadata” under “Data artifacts” on the left side navigation panel
4. On the top tool bar, select “Image/Videos”. A new tab will appear allowing viewership of all files. Again, the goal is to find restricted/confidential images
5. Using the same tool bar, select “Timeline”. This will generate a timeline of all files; when they were created and last accessed. Filter options are on the bottom left panel. This Time graph helps to map out the order of a suspect's actions.

Screenshot Evidence of Claim:

Deleted Files:

sys_case_files - Autopsy 4.19.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing

All

Table Thumbnail Summary

Name	S	C	O	Modified Time	Change Time	Access Time
f0000000_business_Strategy.pdf			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
f0001128_Drilling_Methodology.pdf			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
f0002256.pdf			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
f0002784.jpg			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
f0002808.jpg			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
f0002824.pdf			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00

File Views

File Types

- Deleted Files
- File System (6)
- All (12)

MB File Size

Data Artifacts

Metadata (11)

Analysis Results

Keyword Hits (1)

Single Literal Keyword Search (0)

Single Regular Expression Search (0)

Email Addresses (1)

OS Accounts

Tags

Reports

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

1 of 3 100%

Field 007-A

Field 007-A was recently located in Nigeria. Current discussions with local government have been successful and the below methods will be used in the various parcels of the field. Proprietary methods listed below will result in maximum extraction with little environmental concerns. Current business risk is competitors learning the methodology and steps used in the location based on soil conditions. The company first to deploy these technologies should be able to extract more than 75% of available oil.

Page 1 / 3

Result: 1 of 1 Result	
Type	Value
Version	1.3
Date Modified	2021-08-11 17:41:31 EDT
Date Created	2021-08-11 17:41:31 EDT
Owner	Mike Morris
Source File Path	/img_JSmith_Q1.001/\$CarvedFiles/0001128_Drilling_Methodology.pdf
Artifact ID	-9223372036854775797

Case View Tools Window Help

[Add Data Source](#)
[Images/Videos](#)
[Communications](#)
[Geolocation](#)
[Timeline](#)
[Discovery](#)
[Generate Report](#)
[Close Case](#)

Listing

All

Table Thumbnail Summary

Name	S	C	O	Modified Time	Change Time	Access T
f0000000_business_Strategy.pdf			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-
f0001128_Drilling_Methodology.pdf			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-
f0002256.pdf			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-
f0002784.jpg			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-
f0002808.jpg			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-
f0002824.pdf			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-

File Views

- File Types
- Deleted Files
- File System (0)
- All (12)

Data Artifacts

- Metadata (11)
- Analysis Results
- Keyword Hits (1)
- Single Literal Keyword Search (0)
- Single Regular Expression Search (0)
- Email Addresses (1)

OS Accounts

Tags

Reports

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

1 of 5 69%

Oil company
CONFIDENTIAL Restricted
Business Strategies 2021

Innovating energy solutions: Research and development highlights

R&D has been part of Oil Company since our company began. Our innovations have helped provide the energy fundamental to modern life – from the clean, efficient facts that power today's transportation to the natural gas that provides light and heat to homes and businesses.

Today, our world faces a dual challenge: meeting growing demand for energy while also reducing environmental impacts, including the risk of climate change. Oil Company is

Page 1 / 5

Type	Value
Version	1.3
Date Modified	2021-08-11 18:58:52 EDT
Date Created	2021-08-11 18:58:52 EDT
Owner	Mike Morris
Source File Path	/img_JSmith_Q1.001//CarvedFiles/f0000000_business_Strategy.pdf
Artifact ID	-9223372036854775796

Notice: file is not by john smith but Mike Morris. Furthermore, this is a carved file, indicating that it was intentionally deleted by john smith to avoid being caught.

f0002824.pdf 1 0000-00-00 00:00:00 0000-00-00 00:00:00

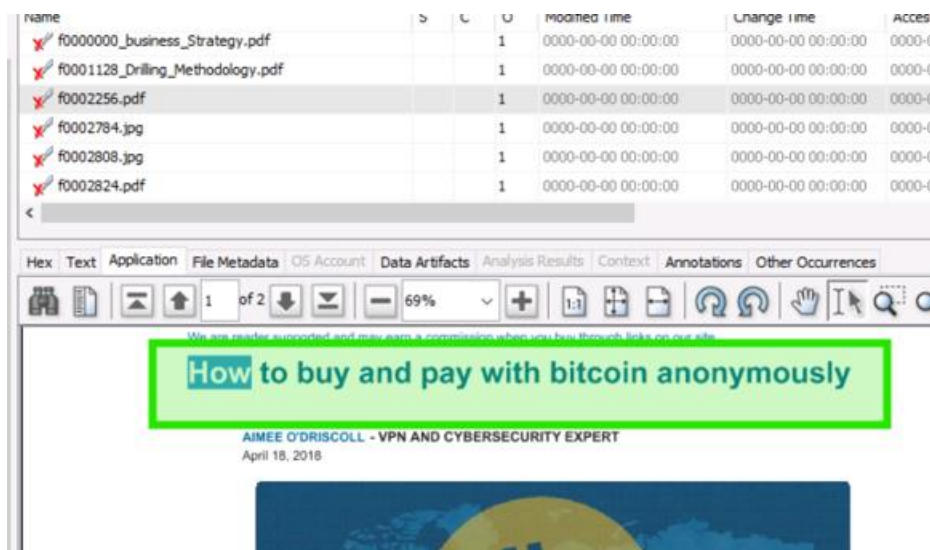
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

1 of 3 58%

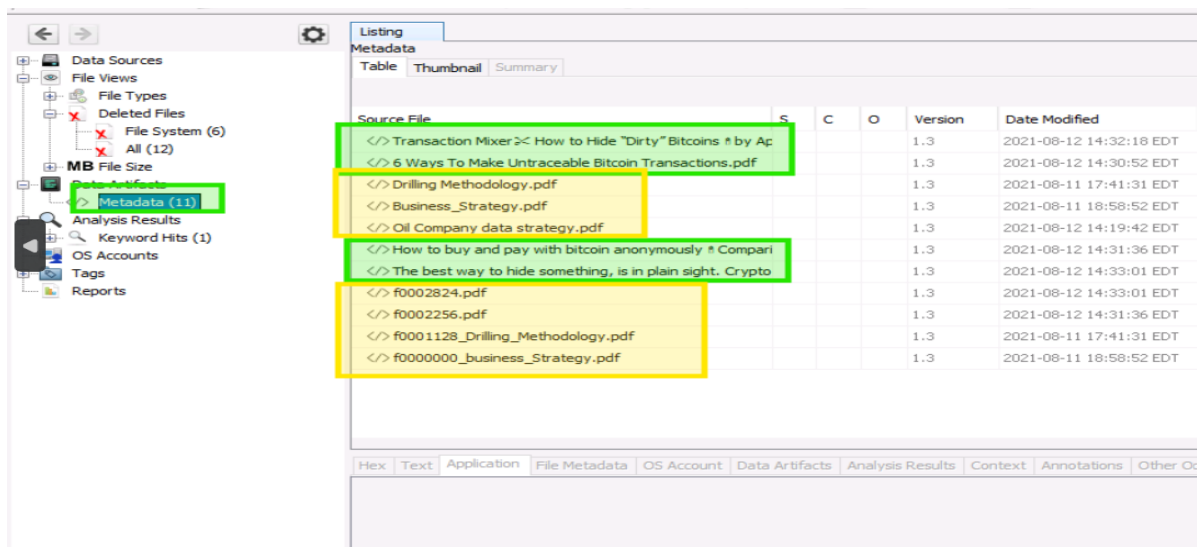
Become a Member # About Us Licensing Advisory Services Deans List

Blockchain courseware

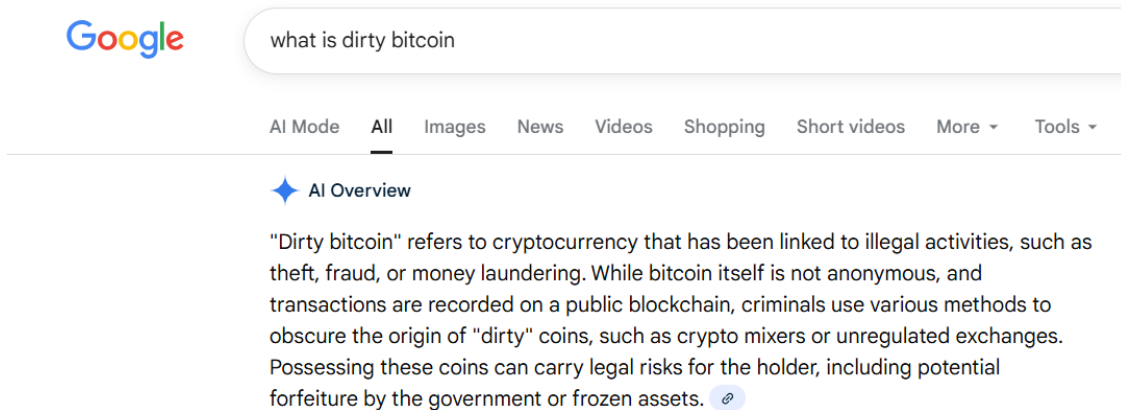
The best way to hide something, is in plain sight. Crypto laundering.



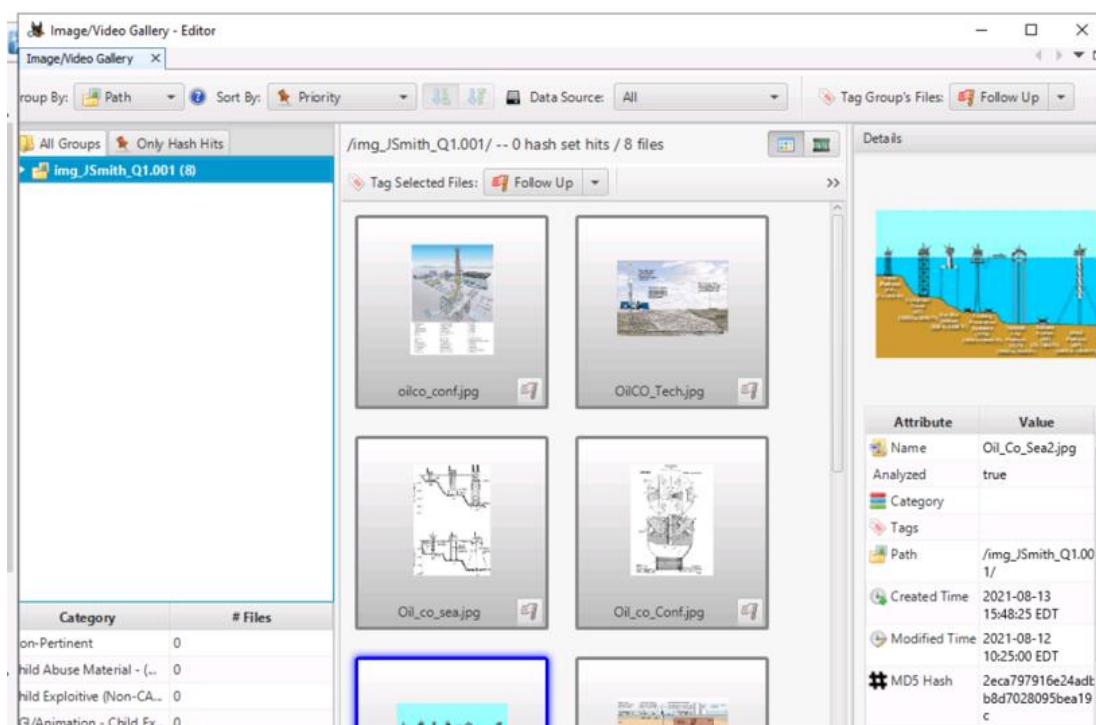
Metadata:



Notice: The metadata marked in yellow is proprietary information that John Smith possesses; green regards john smiths search into bitcoin as a means of anonymous transaction. This more than likely is not a coincidence, and he plans to sell confidential information for bitcoin. Just read the metadata headlines! A simple google search about "dirty bitcoin" is very telling:

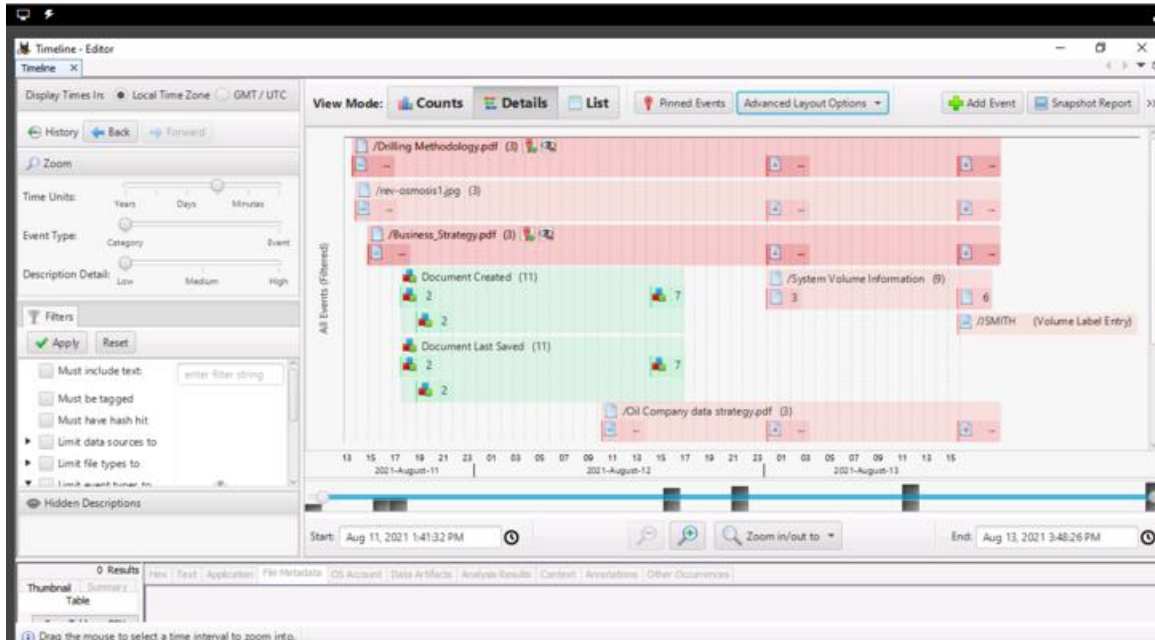


Proprietary images:



Notice: many proprietary images that john smith shouldn't have access to can be found in this screenshot.

Timeline:





Notice: Timeline shows that both proprietary data access and research into bitcoin money laundering occur at the same time. This further implies that john smith not only has accessed confidential information that he didn't have permission to access but his motive was to sell the confidential information to an external party for monetary gain.

3. Findings and Conclusion of Investigation:

Findings:

1. Unauthorized access to confidential information:

- Out of 14 notable pieces of evidence, 7 of them were confidential documents (4 original, 3 copies)
- Name examples: "Business_Strategy.pdf", "Dilling_Methodology.pdf", and "Oil_Company_data_strategy"

2. Evidence of deletion:

- In Autopsy, under file type there are 12 deleted/ carve files, all of which are incriminating

3. Plans regarding data exfiltration:

- Four documents found containing information regarding bitcoin money laundering, hiding 'dirty money', maintain transaction anonymity, and hiding evidence

4. Policy Violations:

- According to the oil company, "All employees sign nondisclosure agreements (NDAs) and acceptable use policies (AUPs)"
- Non-disclosure agreement is violated by accessing confidential information and planning to exfiltrate it
- Obviously, plans of data exfiltration are strictly prohibited by any functional acceptable use policy

Conclusion:

From this forensic analysis, it has been made evident that John Smith accessed, copied, transmitted, and deleted proprietary information (without authorization), and had plans to exfiltrate data from the organization. Digital evidence and artifacts recovered from Autopsy, including proprietary documents, images, and dirty bitcoin transaction documents are supportive evidence of criminal activity, subject to legal and administrative punishment.

References

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